LABORATORY EVALUATION OF PRIMARY HEMOSTASIS

Tests **vascular** and **platelet response**

TEST FOR VASCULAR RESPONSE (INTEGRITY)

- Measures the ability of the platelets to maintain capillary integrity.
- Results correlate with: degree of thrombocytopenia and bleeding time
- Clinical significance (+):
  - ✓ Purpura
  - ✓ Dengue Fever
  - ✓ Scurvy

1. TOURNIQUET TEST

- **Also known as:** Capillary Fragility Test, Capillary Resistance Test, Rumple Leede CFT, Hess Test and Grocco-Frugoni Test
- Uses **POSITIVE PRESSURE TECHNIQUE**
- Principle: By partially obstructing the flow of venous blood, the capillary pressure is increased. This will give rise **to intravasation of blood** which will be manifested in the form of small hemorrhage called *petechiae*
- It tests the stability of the small blood vessels to retain the red cell in their lumen under conditions of stress and trauma.
- Standard pressure: 40-60 mmHg
- Area where petechiae are counted: 1 inch

1. TOURNIQUET TEST

- Important factors:
  - ✓ Platelets
  - ✓ Vitamin C
  - ✓ Collagen

*All are important in the maintenance of normal capillary integrity or resistance.

- A. QUICK'S METHOD: pressure: 40-60 mmHg
- B. GOTHLIN METHOD: pressure: 35-50 mmHg

GRADE	NUMBER OF PETECHIAE	INTERPRETATION
1+	1-10	Few petechiae on the anterior part of the forearm
2+	11-20	Many petechiae on the anterior part of the forearm
3+	21-50	Multiple petechiae over the whole arm and the back of the hand
4+	>50	Confluent petechiae on the anterior part and back of the hand

2. PETECHIOMETER METHOD

- •Also known as: Suction Cup Method
- Uses **NEGATIVE PRESSURE TECHNIQUE** (trauma is created by means of suction cups applied to the skin)
- Principle: It employs the use of a modified **da Silva Melle instrument**. The cup is applied to the outer surface of the arm for a period of 1 minute at 200mmHg
- Normal value: <4 petechiae

PLATELET EVALUATION

QUALITATIVE TESTS: assesses platelet function/morphology

QUANTITATIVE TESTS: assesses platelet count

QUANTITATIVE PLATELET TEST:

- part of coagulation workup/profile

- **PLATELET COUNT:** valuable in assessing bleeding disorders associated with thrombocytopenia, uremia, liver disease, or malignancies, and in monitoring the course of disease among patients with bone marrow failure

THROMBOCYTOPENIA

- Platelet count is BELOW the normal reference range
- diseases include:

  - **Marrow infiltrative processes** (ex: Leukemia, Myelofibrosis, Multiple Myeloma, Metastatic carcinoma)
  - **Aplasia** (ex: Aplastic anemia, Fanconi anemia, chemotherapy, and radiation
  - **Ineffective blood production** (ex: Vitamin B12 deficiency, Folic Acid deficiency, alcohol ingestion, myelodysplastic syndrome and PNH)
  - **Congenital disorders** (ex: Bernard Soulier Syndrome and May-Hegglin anomaly)

2 Main Reasons of Thrombocytopenia

1. ABNORMAL PLATELET DISTRIBUTION

Hypersplenism

Kasabach-Merit Syndrome

2. INCREASED PLATELET DESTRUCTION

<u>Primary Immune Disorders</u>

-Immune Thrombocytopenic Purpura

-Post-transfusion Purpura

-Heparin-induced thrombocytopenia

2 Main Reasons of Thrombocytopenia

2. INCREASED PLATELET DESTRUCTION

Secondary Immune Disorders

-Systemic Lupus Erythematosus

-Lymphoproliferative disorders

-Viral infection

Microangiopathic Thrombocytopenia

-Thrombotic Thrombocytopenic Purpura

-Disseminated Intravascular Coagulation

-Hemolytic Uremic Syndrome

CAUSES OF THROMBOCYTOPENIA

CONDITIONS	EXAMPLES
MARROW INFILTRATIVE PROCESS	✓ LEUKEMIA ✓ MYELOFIBROSIS ✓ MULTIPLE MYELOMA ✓ METASTATIC CARCINOMA ✓ LYMPHOMA
APLASIA	✓ APLASTIC ANEMIA ✓ FANCONI ANEMIA ✓ CHEMOTHERAPY ✓ RADIATION
INEFFECTIVE BLOOD PRODUCTION	✓ VITAMIN B12 DEFICIENCY ✓ FOLIC ACID DEFICIENCY ✓ ALCOHOL INGESTION ✓ MYELODYSPLASTIC SYNDROME ✓ PNH
CONGENITAL DISORDERS	✓ BERNARD-SOULIER SYNDROME ✓ MAY-HEGGLIN ANOMALY
ABNORMAL PLATELET DISTRUBUTION	✓ HYPERSPLENISM ✓ KASABACH-MERITT SYNDROME
INCREASED PLATELET DESTRUCTION	✓ PRIMARY IMMUNE ✓ SECONDARY IMMUNE ✓ MICROANGIOPATHIC

THROMBOCYTOSIS

- Platelet count is ABOVE the normal reference range
- diseases include:

1. Myelodysplastic syndromes

- ✓ Polycythemia vera
- ✓ Essential thrombocytosis

2. Reactive thrombocytosis

- ✓ After hemorrhage or surgical procedure
- ✓ Iron Deficiency Anemia
- ✓ Epinephrine administration
- ✓ Malignancies

CAUSES OF THROMBOCYTOSIS

MYELODYSPLASTIC SYNDROMES	✓ POLYCYTHEMIA VERA ✓ ESSENTIAL THROMBOCYTHEMIA
REACTIVE THROMBOCYTOSIS	✓ AFTER HEMORRHAGE OR SURGICAL PROCEDURE ✓ IRON DEF. ANEMIA ✓ MALIGNANCIES ✓ EPINEPHRINE ADMIN.

Challenges in platelet count:

- Small size: 2-4um
- Adheres to foreign surfaces (skin and walls of pipet)
- Easily disintegrate
- Hard to differentiate from debris
- Unevenly distributed in the blood since they tend to clump

3 WAYS TO QUANTIFY PLATELETS

1) PLATELET ESTIMATION USING PBS

INDIRECT PLATELET COUNT METHOD

- may detect PLATELET CLUMPING and CLOTS from poorly collected specimens
- NOT RELIABLE because the results depend upon the distribution of platelets and on the red cell count
- Platelets are counted in their relationship to RBCs on a fixed-stained smear

ADEQUATE PLATELET COUNT is reported when:
• 1 platelet per 10-40 RBC • 3-10 platelets per 100 RBC (or in 1 OIO field) • 5-20 platelets per 200 RBC • 8-20 platelets per field (in 10 OIO fields)

INDIRECT PLATELET COUNT

1. FONIO'S METHOD

- Uses: 14% Magnesium Sulfate + 1 drop of blood +
- Wright stain

2. DAMESHEK METHOD

- Uses: Brilliant Cresyl Blue, sodium citrate, sucrose and formalin; then counterstained with Wright's stain
- Platelets/uL= $\frac{\text{plts counted x RBC count}}{1000}$
- Reference Range: 250-500 x 109/L or 250,000-500,000

INDIRECT PLATELET COUNT

3. MODIFIED INDIRECT METHOD

- Number of platelets are counted per field
- ADEQUATE PLATELET COUNT: 8-20 platelets/field
- Number of fields counted: 10

- Platelets/uL= $\frac{\text{plts counted in 10 fields x 20,000}}{10}$
- Reference range: 150-350 x 109/L or 150,000-350,000

PLATELET COUNT	ESTIMATION
0-49,000/uL	Markedly decreased
50,000-99,000/uL	Moderately decreased
100,000-149,000/uL	Slightly decreased
150,000-199,000/uL	Low normal
200,000-400,000/uL	Normal
401,000-599,000/uL	Slightly increased
600,000-800,000/uL	Moderately increased
>800,000/uL	Markedly increased

3 WAYS TO QUANTIFY PLATELETS

2) ABSOLUTE PLATELET COUNT USING HEMOCYTOMETER

DIRECT PLATELET COUNT METHOD

a. REES AND ECKER METHOD

- Also known as TONCANTIN'S METHOD
- Uses Rees-Ecker diluting fluid (storage temp: 0-4'C)
- Components of diluent: Distilled Water, Formalin Buffer and Brilliant Cresyl Blue Uses **RBC pipette (1:200)** or WBC pipette (1:20)

b. GUY AND LEAKE

- Reagent contains: Formalin Buffer & Crystal Violet

3 WAYS TO QUANTIFY PLATELETS

2) ABSOLUTE PLATELET COUNT USING HEMOCYTOMETER

DIRECT PLATELET COUNT METHOD

c. BRECKER-CRONKITE

- Uses: Ammonium Oxalate
- Microscope: Phase Contrast Microscope

d. UNOPETTE SYSTEM

- Diluting fluid: Ammonium oxalate
- Dilution: 1:100
- It uses the entire RBC squares for counting platelet
- Platelet count formula: number of platelets X 1000

DIRECT PLATELET COUNT

REES AND ECKER	• Diluent: sodium citrate, formalin buffer, Brilliant Cresyl Blue, distilled water • Uses: LIGHT MICROSCOPE • Platelets appear as highly refractile round bodies • **Plts/uL= $\frac{\text{plts counted x 10 x 200}}{4}$**
GUY AND LEAKE	• Diluent: sodium oxalate, 40% formalin, and crystal violet • Uses: LIGHT MICROSCOPE • **Plts/uL= plts counted x 5 x 10 x 100**
BRECKER-CRONKITE	• Diluent: 1% Ammonium oxalate • Uses: PHASE CONTRAST MICROSCOPE • MOST ACCURATE METHOD • **Plts/uL= plts counted x area correction factor x 10 x 100**
UNOPETTE	• Diluent: 2% Ammonium oxalate • Uses: Bright field microscopy • It uses the entire RBC squares for counting platelet • Platelet count formula: number of platelets X 1000

Reference Range: 150-400 x 109/L or 150,000-400,000/uL

3 WAYS TO QUANTIFY PLATELETS

3. USE OF AUTOMATED COUNTING MACHINE

- The automated machine determines the platelet count depending on the average volume of the cell.
- RBC and Platelet are counted simultaneously by the machine and the machine sorted these 2 cells depending on their size or volume.
- Normal platelet volume is approximately **7 fL**
- If the volume falls within **2-20 fL** it will be registered by the machine as platelet

3. AUTOMATED/ELECTRONIC METHOD

- The automated machine determines the platelet count depending on the average volume of the cell.
- RBC and Platelet are counted simultaneously by the machine and the machine sorted these 2 cells depending on their size or volume.
- Normal platelet volume is approximately **7 fL**
- If the volume falls within **2-20 fL** it will be registered by the machine as platelet
- Counts platelets either by: *voltage-pulse counting* or *electro-optical counting*

SOURCES OF ERROR

FALSELY INCREASED COUNT

- ✓ Fragments of leukocyte cytoplasm
- ✓ Inclusions from lysed RBCs
- ✓ Debris

FALSELY DECREASED COUNT

- ✓ Platelet satellitism
- ✓ Platelet clumping due to agglutinins
- ✓ Spontaneous platelet aggregation
- ✓ Clotting due to faulty blood collection and handling
- ✓ Delay in sampling

QUALITATIVE PLATELET TESTS:

Tests for specific functions of platelet

- ✓ Adhesion
- ✓ Aggregation
- ✓ Clot retraction
- ✓ Factor assays

TEST OF PLATELET ADHESION

BLEEDING TIME

- Time required for a standardized wound to stop bleeding
- Standard skin wound: 2-3mm/2-4mm
- Tests the in **vivo function of platelets** and number
- Determines both congenital and acquired platelet disorder
- Factors affecting bleeding time:
  - ✓ Elasticity of tissue
  - ✓ Ability of the blood vessels to constrict and retract
  - ✓ Mechanical and chemical action of platelets in the formation of hemostatic plug

BLEEDING TIME

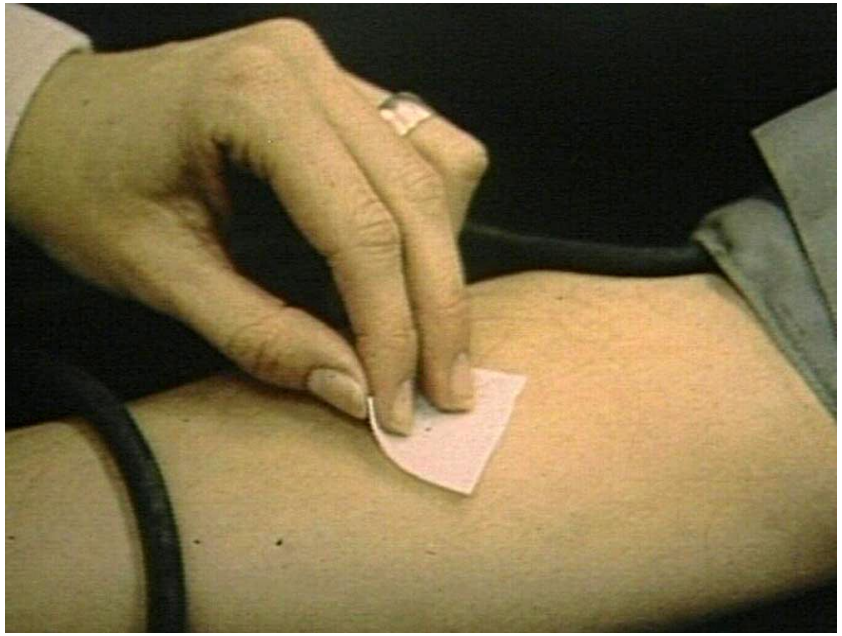
A. DUKE'S METHOD

- ✓ A standard wound is made on various site (ex. fingers toes, and earlobes)
- ✓ Site of choice: fingers
- ✓ Reference range: 2-6 minutes

B. MODIFIED IVY'S METHOD

- ✓ BEST SCREENING TEST available for platelet's role in hemostasis
- ✓ 2-3 **incisions** are made on the VOLAR SURFACE of the patients forearm with application of pressure usually around 40mmHg during the test
- ✓ Reference range: 2-9 minutes

MODIFIED IVY'S METHOD



BLEEDING TIME

C. MODIFIED SIMPLATE METHOD

- ✓ Modification of the Ivy's method
- ✓ It uses a **spring-loaded lancet** that will create exactly a standard skin wound on the chosen site
- ✓ Standard skin wound: 5mm x 1mm

D. COPLY LALITCH METHOD

- ✓ It involves puncturing the finger (up to 6mm deep) and immersion of the punctured finger in sterile physiologic saline solution warmed at 37°C
- ✓ Reference range: >3 minutes

E. ADELSON-CROSBY METHOD

- ✓ Same with Coply-Lalitch method

BLEEDING TIME

F. MACFARLANE's METHOD

- ✓ Same principle with Adelson-Crosby but uses **earlobe** as the site puncture

G. ASPIRIN TOLERANCE TEST

- ✓ Assesses the effect of a standard dose of aspirin on the Duke's bleeding time

BLEEDING TIME IS PROLONGED IN THE FF. CONDITIONS:

1. **When the blood platelets are greatly reduced**
   a. Thrombocytopenic purpura- dec. In platelets
   b. Acute Leukemia
   c. Aplastic anemia
2. **Injury of capillary wall**
   a. Scurvy
   b. Toxins (infection, chemical, snake venom)
3. **Plt deficiency**
   a. Destructive disease of the liver
   b. HDN
4. **Slightly prolonged severe anemia**

ABNORMAL BLEEDING TIME IS OBSERVED IN:

- ✓ Thrombocytopenia
- ✓ Hypofibrinogenemia
- ✓ vWF disorder
- ✓ Connective tissue disorders
- ✓ Scurvy
- ✓ Bernard Soulier Syndrome

PLATELET RETENTION TEST

- Principle: The **adhesiveness property of platelets** maybe assessed and measured by their ability to adhere on glass surfaces with beads
- Anticoagulant of Choice: EDTA
- Normal value: 75%-95% of platelets should adhere

***Salzmann Method:** tests for the retention of platelets within the glass bead column

ABNORMAL PLATELET RETENTION TEST

DECREASED	INCREASED
✓ Bernard-soulier syndrome	✓ Thrombotic disorders
✓ Glanzmann’s thrombasthemia	✓ Hyperlipidemia
✓ Vwf DISEASE	✓ Carcinoma
✓ Chediak-higashi	✓ Oral contraceptives
✓ Myeloproliferative disorders	✓ Pregnancy
✓ Uremia	
✓ Aspirin admin.	

TESTS FOR PLATELET AGGREGATION

- ✓ Assess the ability of platelets to aggregate after addition of an aggregating agent
- ✓ Anticoagulant of choice: CITRATE
- ✓ Sample required: **PLATELET-RICH PLASMA**
- ✓ Initially the PRP is turbid due to the presence of platelets then by the addition of aggregating agents, *the platelet will then change its shape and later aggregate which decreases the turbidity of the specimen.*
- ✓ The decrease in the turbidity of the PRP is measured by the **platelet aggregometer**

AGGREGATING AGENTS USED

- ✓ ADENOSINE DIPHOSPHATE (ADP)
- ✓ EPINEPHRINE
- ✓ THROMBIN
- ✓ ARACHIDONIC ACID
- ✓ RISTOCETIN
- ✓ COLLAGEN
- ✓ SEROTONIN

TEST CONSIDERATIONS

- ✓ No hemolysis
- ✓ Fasting: 8 hours
- ✓ pH: 6.5-8.5
- ✓ Temperature: 37°C
- ✓ Patient should not take any NSAIDS at least 7 hours before the test
- ✓ Performed within 3 hours

INTERFERING FACTORS:

- ✓ Platelet count: < 100,000/uL
- ✓ Lipemia will interfere with testing
- ✓ 10 days before test, drugs that inhibit platelet aggregation are contraindicated (ex. aspirin, antihistamines, steroids, cocaine, anti-inflammatories, theophylline, and antibiotics)
- ✓ On the day of the test, avoid caffeine
- ✓ Avoid warfarin/coumadin for 2 weeks and heparin therapy for 1 week before testing

TEST FOR CLOT RETRACTION

- Measure the entire function of platelets
- It also detects clotting factor deficiencies
- Depends on the normal levels of:
  - ✓ Fibrinogen
  - ✓ Calcium
  - ✓ Atp
  - ✓ Platelets

METHODS OF CLOT RETRACTION	
QUALITATIVE	**QUANTITATIVE**
✓ HIRSHBOECK METHOD -also known as :CASTOR OIL METHOD	✓ STEFANINI-DAMESHEK ✓ MCFARLANE METHOD

PLATELET FACTOR ASSAYS

1.PF3 AVAILABILITY

- Platelet factor 3 is needed for the continuous activation of the clotting process
- This test uses PPP, PRP and activators such as kaolin or elagic acid to activate solely the platelets and the clotting of both specimens are determined.
- The clotting time of PRP incubated with kaolin and epinephrine is considerably shorter than that of platelet poor plasma.

2. PF4 AND B-TG Assay

- Their presence in vivo indicates **on going platelet activation**
- RIA is the method used for this assay

- End of presentation